BorNEO Hackwknd 2026

Case Study 5: Disaster Resilience AI

Team **HACHIMI**

UNIVERSITI TEKNOLOGI MALAYSIA (UTM)

Team Leader	ALFRED CHIN ZHAN HOONG
Team Member 1	GUI GUANYU
Team Member 2	LEONG JUN QIANG
Team Member 3	NG JAA WEI

1. Introduction

Background of Case Study

With over 5,000 disasters recorded in the ASEAN region since 2012, Southeast Asia faces an urgent and escalating need for enhanced disaster preparedness. The ASEAN Capacity Building Roadmap 2025–2030 on Disaster Management outlines a regional shift toward proactive safety culture through technology adoption, capacity building, and community-level resilience. Rural populations in Borneo — spanning Malaysian states like Sabah and Sarawak, and Indonesian Kalimantan — are particularly at risk. These communities often lack precise geolocation infrastructure, real-time hazard data, and structured emergency coordination needed for rapid rescue during monsoon floods, landslides, and tidal surges. BorNEO AI was developed in direct response to this challenge: a web-based, AI-driven disaster resilience platform that connects residents, responders, and administrators through one unified system for medical identity, localised alerts, AI-powered preparedness education, and SOS coordination.

Chosen SDG

SDG 11: Sustainable Cities and Communities — Make cities and human settlements inclusive, safe, resilient, and sustainable.

BorNEO AI directly supports:

- Target 11.5 - Significantly reduce the number of deaths and people affected by disasters, with a focus on protecting the poor and people in vulnerable situations.
- Target 11.b - Substantially increase the number of cities and human settlements adopting and implementing integrated policies and plans towards inclusion, resource efficiency, mitigation and adaptation to climate change, and resilience to disasters, in line with the Sendai Framework for Disaster Risk Reduction 2015–2030.

This aligns with the ASEAN Capacity Building Roadmap 2025–2030's emphasis on strengthening local-level preparedness and leveraging AI for disaster risk reduction across Member States.

Problem Statement

During floods, landslides, and tidal surges in Borneo and the broader ASEAN region, field teams lose valuable time identifying victims, locating households, and understanding medical risk. Rural populations often lack precise coordinates for their homes, making rescue operations slower and less effective. Existing solutions are fragmented — emergency contacts live in phone books, medical data is on paper, alert systems are one-directional, and preparedness education is generic rather than personalised to local hazards, household composition, or medical vulnerabilities. There is no single platform that ties together crisis preparation, real-time situational awareness, and rapid-response coordination for both residents and responders in a way that serves ASEAN's rural communities.

Objectives

1. Proactive preparedness through AI-driven e-learning — Deliver specialised disaster management training through interactive AI-generated survival simulations and personalised emergency checklists, aligning with the ASEAN Capacity Building Roadmap's goal of equipping local communities to respond to and recover from major events.
2. Hyper-local early warning and real-time monitoring — Provide centralised, multilingual alert broadcasting with map-based situational awareness and severity-based filtering for floods, landslides, and tidal hazards.
3. Rapid rescue with precise coordinates and AI triage — Enable SOS submission with automatic GPS capture, AI-assisted severity classification, and digital medical rescue cards with QR codes, addressing the critical gap in precise coordinate data for rural rescue operations.
4. Role-based workflows for responders and administrators — Provide a dedicated workspace for SOS queue management, AI triage review, alert creation, and dispatch coordination.

2. Review of Existing Solutions

Solution	Strengths	Gaps / Limitations
MySejahtera (MY)	Wide adoption; centralised health data	Not designed for disaster response; no SOS, rescue-card, or AI triage features
FEMA App (US)	Real-time alerts; safety tips	US-centric; no personalised checklists, no medical identity, no AI triage
Banjir.info (MY)	Crowd-sourced flood data	Read-only flood map; no preparedness tools, no responder workflow, no rescue coordination
Red Cross Emergency App	Multi-hazard alerts; first-aid guidance	Generic global content; no localised Borneo/ASEAN context, no rescue-card QR system
WhatsApp / Social Media	Ubiquitous; real-time communication	Unstructured data; misinformation risk; no role-based triage, no AI classification, no GPS-precise SOS

Key gaps BorNEO AI addresses:

- No existing platform combines medical identity (rescue cards + QR), AI-assisted SOS triage, personalised preparedness education, and admin alert broadcasting in one system tailored to ASEAN's rural context.
- Most solutions are one-directional — they push alerts but don't let residents send structured distress signals with precise GPS coordinates back to responders.
- Preparedness content is generic globally; none generates checklists personalised to household size, pets, medical conditions, and specific hazard type for a Southeast Asian monsoon/tropical context.
- No solution provides AI-driven e-learning (survival simulations) specifically designed for ASEAN hazard scenarios, as called for in the ASEAN Capacity Building Roadmap 2025–2030.

3. Methodology

Technical Track

i) Tech Stack & Architecture:

Layer	Technology
Frontend	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS 4, MUI Components
Backend	Next.js Route Handlers (serverless API routes)
Database	PostgreSQL (Supabase) via Prisma ORM with @prisma/adapter-pg
Authentication	Supabase Auth (magic-link passwordless login)
AI Engine	Google Gemini 2.5 Flash via @google/generative-ai
Maps & Geocoding	Google Maps JavaScript API, Geocoding / Reverse Geocoding APIs, @vis.gl/react-google-maps
Utilities	qrcode for rescue-card QR generation

ii) Architecture:

The application follows a monolithic Next.js App Router structure with clear separation:

- App routes - Resident pages (/, /page-alerts, /page-resources, /page-profile) and admin pages (/admin/sos, /admin/alert).
- API routes - RESTful handlers under /api/ for auth sync, user management, rescue cards, SOS reports, alerts, checklists, geocoding, and three AI endpoints (draft-alert, generate-plan, quiz).
- Prisma data model - Five core entities: User, RescueCard, IncidentReport, Alert, and EmergencyPlan.

iii) Implementation process:

1. Defined the Prisma schema with PostgreSQL, modelling users, rescue cards, incident reports, alerts, and emergency plans.
2. Built Supabase magic-link authentication with a sync route to bootstrap user records into the database.
3. Implemented three core pillars mapped to the hackathon topic:
    - AI-driven e-learning (Pillar 1) - survival simulations via /api/ai/quiz and personalised checklists via /api/ai/generate-plan, addressing the ASEAN Roadmap's call for specialised disaster training.
    - Hyper-local early warning (Pillar 2) - alert broadcasting with map-based monitoring, severity filtering, and AI-drafted multilingual alerts via /api/ai/draft-alert.

    - Rapid rescue with precise coordinates (Pillar 3) - SOS submission with GPS auto-capture, AI triage via /api/reports, rescue cards with QR codes, and home coordinate mapping — directly addressing the gap in precise location data for rural rescue.
4. Integrated Gemini 2.5 Flash across four AI workflows with structured JSON output (responseMimeType: "application/json") for deterministic parsing.
5. Implemented role-based routing so residents and administrators see different navigation and tooling.

iv) AI Integration Details (4 endpoints):

Endpoint	AI Function	Alignment with ASEAN Roadmap
/api/reports	Classifies inbound SOS reports by severity, tags, action, and required assets	Strengthens rapid-response coordination
/api/ai/draft-alert	Converts raw hazard notes into structured trilingual broadcasts (EN/BM/ZH)	Improves communication to multilingual ASEAN communities
/api/ai/generate-plan	Generates personalised 7–10 item emergency checklists based on hazard, family size, pets, and special needs	Delivers specialised disaster management capacity building
/api/ai/quiz	Generates localised survival scenarios with 3 action choices and explanations	AI-driven e-learning for disaster preparedness training

Non-Technical Track

i) Design Thinking Process:

- Empathise: Identified that rural Borneo residents, elderly caregivers, and field responders face fragmented emergency information during monsoon season. School-aged populations lack structured disaster education beyond generic pamphlets.
- Define: The core problem is the absence of a unified digital lifeline that works before, during, and after a disaster — specifically one that provides precise coordinates and AI-assisted decision support for ASEAN's rural context.
- Ideate: Structured the solution around three operational pillars (Prepare, Monitor, Respond), each mapping to a distinct user need and a specific gap identified in the ASEAN Capacity Building Roadmap.

- Prototype: Built a functional web prototype with resident and admin flows, prioritising mobile-responsive layouts and clear emergency UX (e.g., the SOS modal with hazard selection, GPS broadcasting notice, and submit-state feedback).
- Test: Iterated on SOS submission flow (adding 6 hazard types: Flood, Landslide, Tidal, Medical, Infrastructure, Other), modal sizing, and form validation based on internal team testing.

ii) User Flow:

- Residents: Sign in via magic link → complete profile and region → set up rescue card with medical data and home coordinates → explore AI survival simulations and generate personalised checklists → monitor alerts → send SOS with GPS if an incident occurs.
- Administrators: Access admin workspace → review incoming SOS queue with AI triage metadata (severity, tags, dispatch recommendation) → open resident rescue-card data → create or AI-draft regional alerts → coordinate response with location and medical context attached.

4. Results

Objectives Met

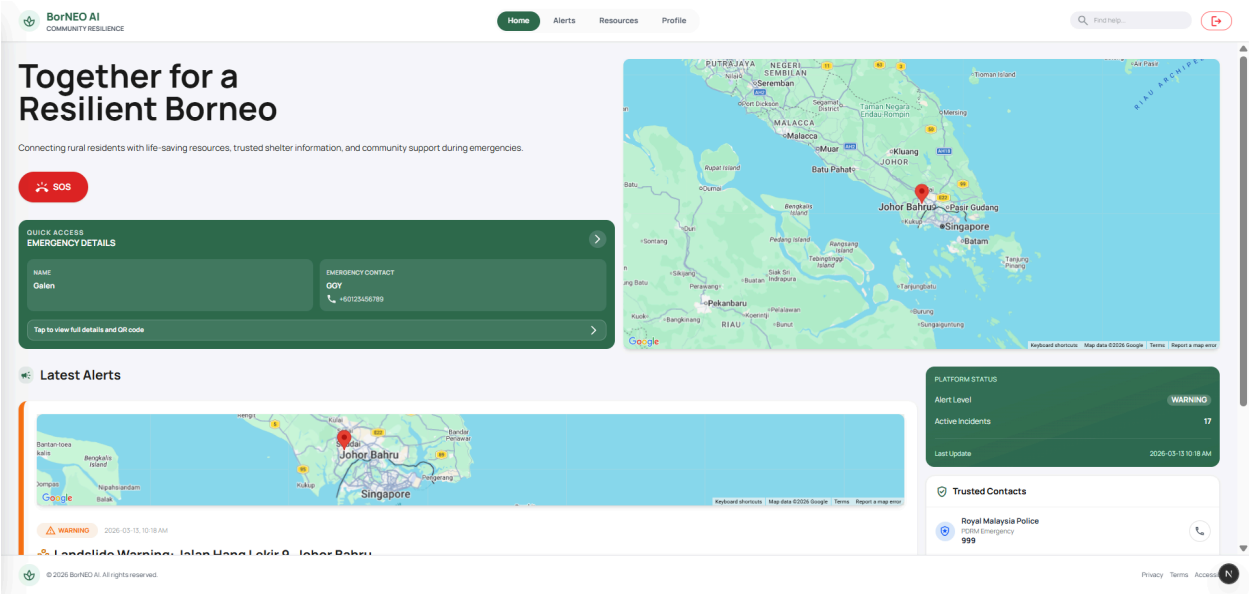
Objective	Status	Evidence
AI-driven e-learning for preparedness	Achieved	AI Survival Simulator with 3 hazard types (Flood, Landslide, Tidal); AI-generated personalised checklists with persistence and completion tracking
Hyper-local early warning	Achieved	Alert feed with list and map views; severity-based filtering (Priority, Warning, Monitor); trilingual AI-drafted broadcasts; Google Maps integration with alert pins
Rapid rescue with precise coordinates	Achieved	SOS submission with 6 hazard types, automatic GPS capture, AI triage (severity, tags, actions, required assets); digital rescue card with QR code; home coordinate storage with map picker
Role-based responder workflows	Achieved	Separate resident/admin navigation; admin SOS queue with triage metadata; admin alert creation with AI draft assistance

Key Outcomes

- SOS reports are automatically geocoded (Google Maps Reverse Geocoding), triaged by AI (Gemini), and persisted to PostgreSQL — providing responders with structured, actionable intelligence instead of raw text. This directly addresses the ASEAN region's need for precise coordinate data during rescue operations.
- Rescue cards store blood type, allergies, medical conditions, emergency contacts, and home coordinates with a generated QR code for field scanning — enabling responders to retrieve critical medical data instantly.
- Emergency checklists are personalised to hazard type, family size, pets, and special needs, then saved to the database for later review and completion tracking — delivering the specialised disaster management training capacity the ASEAN Roadmap calls for.
- Survival simulations generate contextualised Borneo-region scenarios with correct/incorrect answer explanations — functioning as an AI-driven e-learning system for disaster preparedness.
- Alert broadcasting supports AI-assisted trilingual drafting (English, Bahasa Melayu, Chinese) — addressing ASEAN's multilingual communication needs during emergencies.

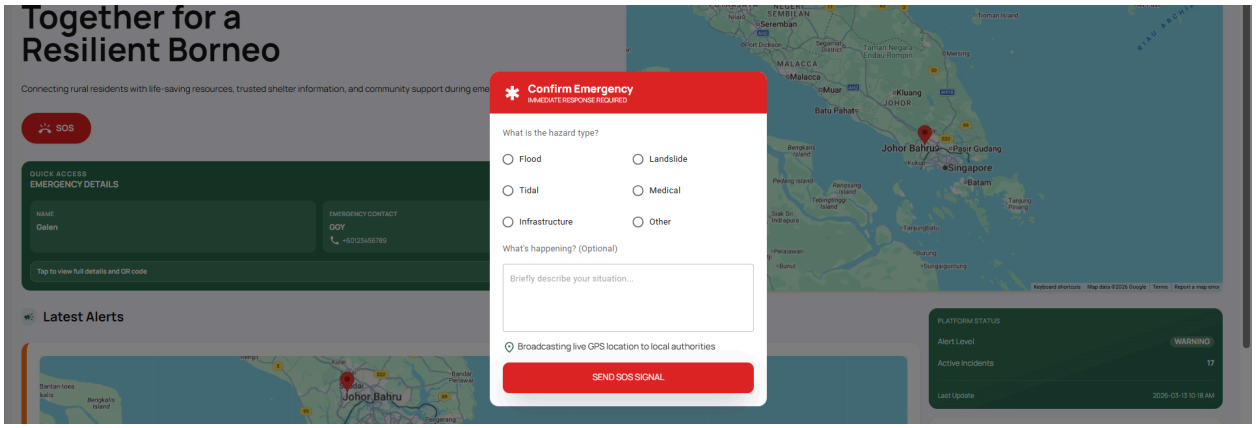
5. Prototype Showcase

Screenshot 1: Resident Home Dashboard



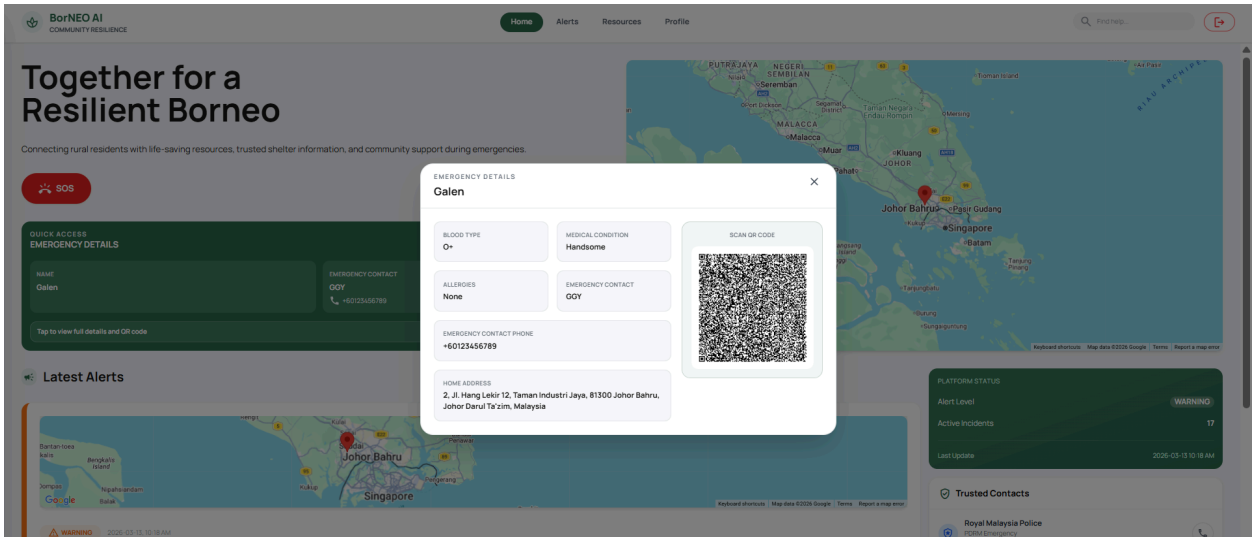
The landing page displays the tagline "Together for a Resilient Borneo", a red SOS button for emergency submission, the user's Emergency Details card (name, emergency contact), a live Google Maps view showing current location, the latest alerts panel with severity-coded cards, platform status, and trusted contacts.

Screenshot 2: SOS Emergency Modal



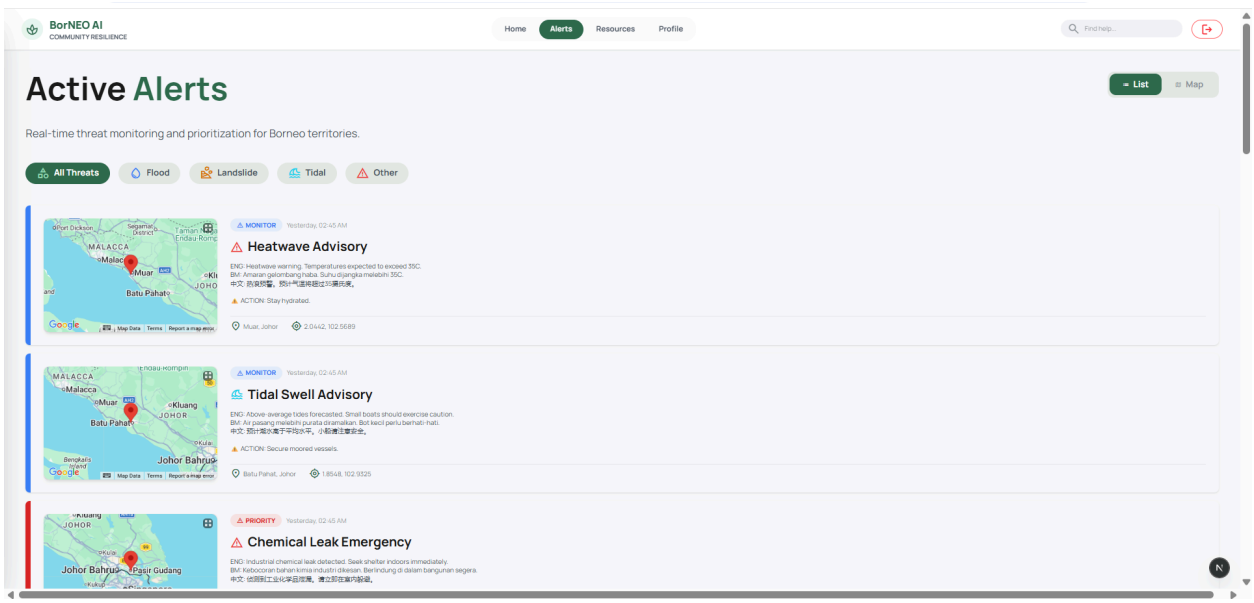
Clicking the SOS button opens a modal titled "Confirm Emergency — IMMEDIATE RESPONSE REQUIRED". Users select from six hazard types (Flood, Landslide, Tidal, Medical, Infrastructure, Other), optionally describe the situation, and see a GPS broadcast notice. The "SEND SOS SIGNAL" button submits with loading state feedback and disables until a hazard is selected.

Screenshot 3: Emergency Details Popup



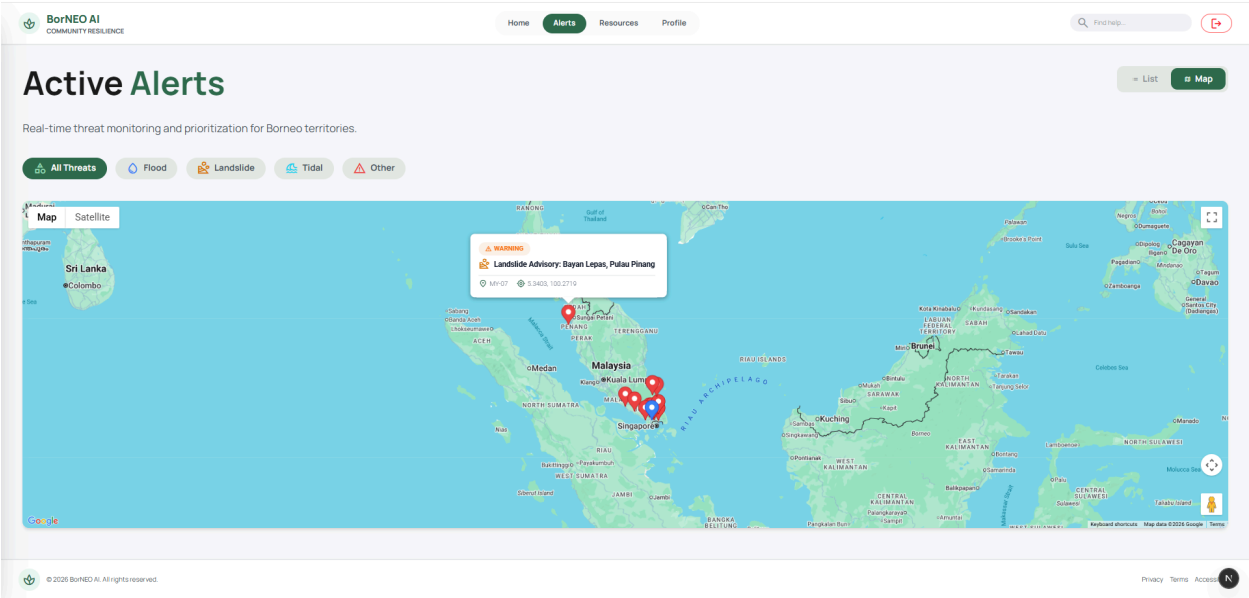
Clicking the Emergency Details card opens a content-fitted modal showing full rescue-card information: name, blood type, medical conditions, home address, emergency contact name and phone, plus a scannable QR code for field responders.

Screenshot 4: Alerts Page — List View



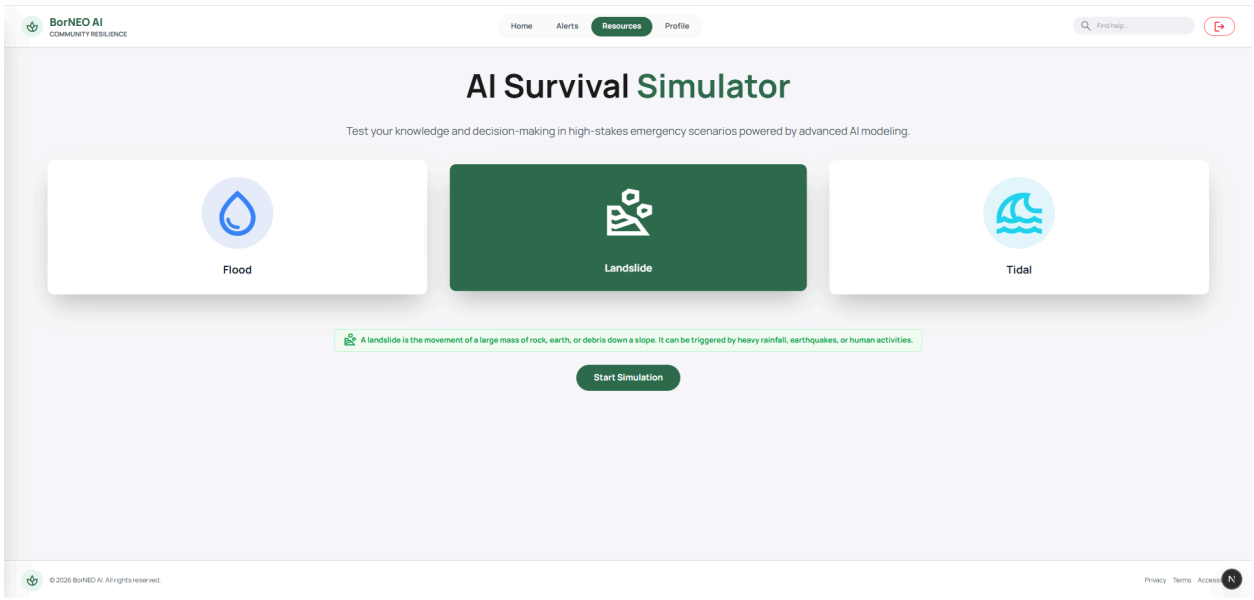
Displays all active alerts with severity-coded cards (Priority / Warning / Monitor), each showing a mini map, hazard category badge, location, timestamp, and multilingual description body. Filter chips allow narrowing by All Threats, Flood, Landslide, Tidal, or Other.

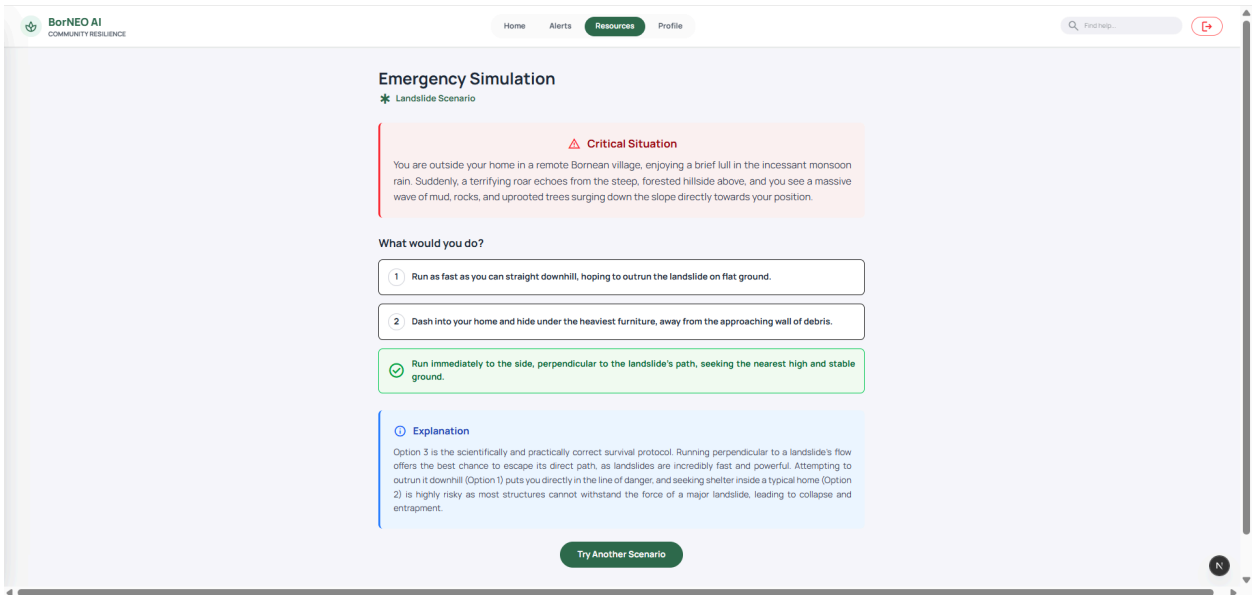
Screenshot 5: Alerts Page — Map View



A full Google Maps view with colour-coded pins for each alert. Hovering a pin shows an info window with alert details. Toggle between List and Map modes via the mode switcher.

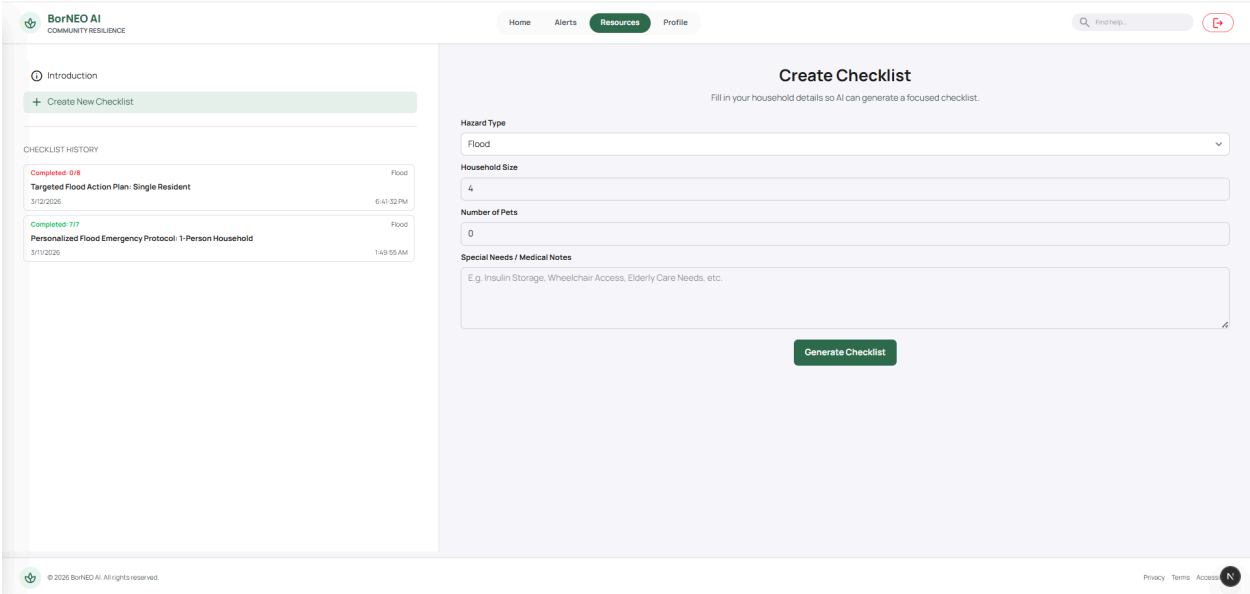
Screenshot 6: AI Survival Simulator

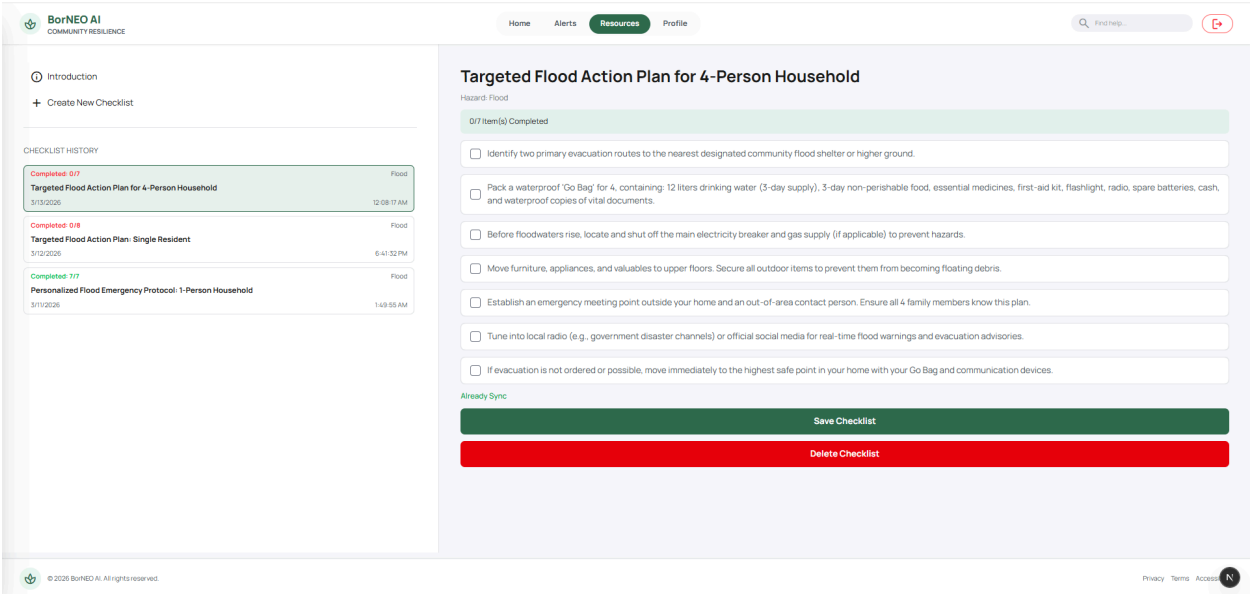




The Resources section includes an AI Survival Simulator where users pick a hazard type (Flood, Landslide, Tidal), then receive a Gemini-generated Borneo-context scenario with three action choices. After selecting, the correct answer is revealed with an educational explanation — functioning as the AI-driven e-learning system aligned with the ASEAN Roadmap.

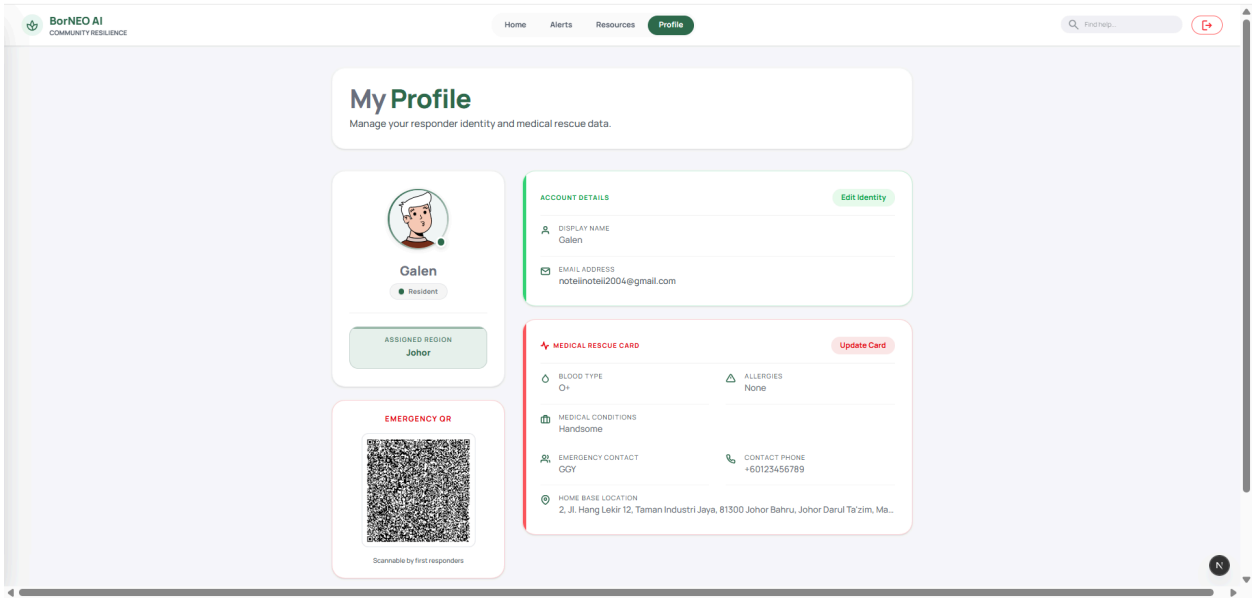
Screenshot 7: Smart Emergency Checklist Generator

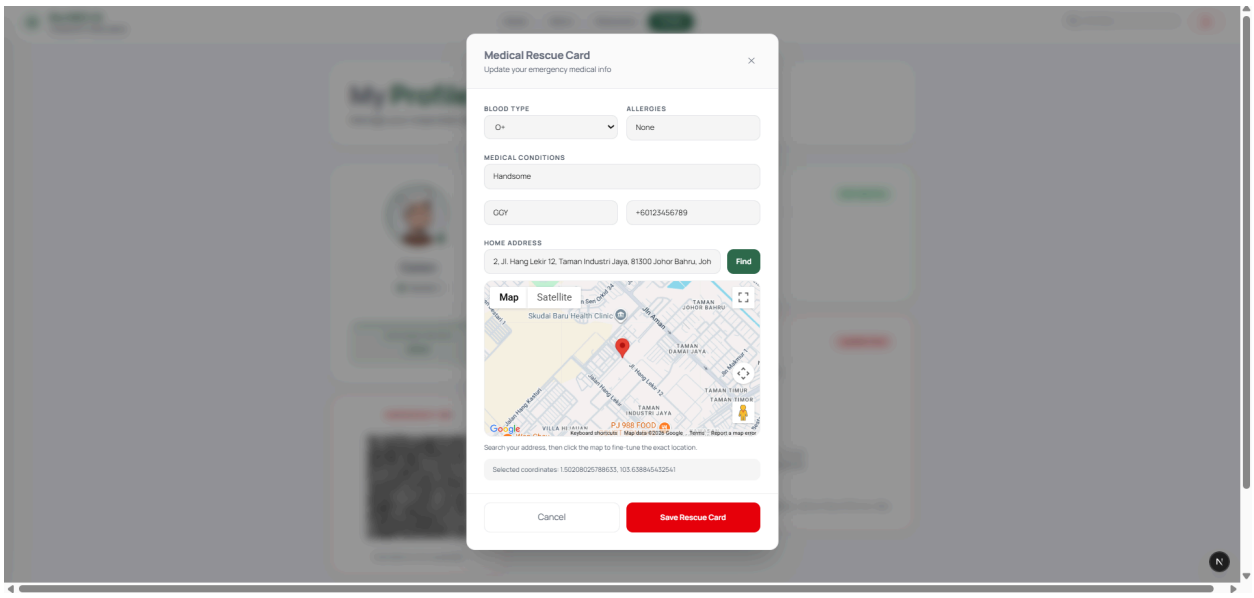




Users input hazard type, family size, pets, and special needs. Gemini generates a tailored 7–10 item emergency checklist (e.g., "Pack 12 litres of water — 3-day supply for 4 people"). Checklists are saved to the database and items can be checked off over time.

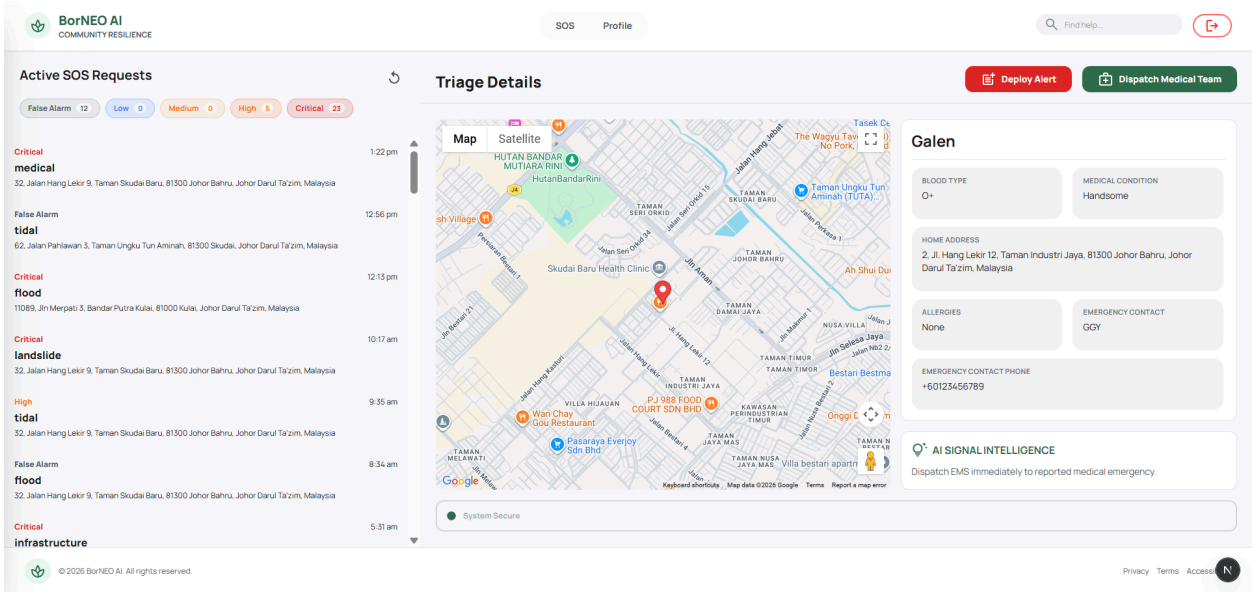
Screenshot 8: Profile & Rescue Card Management

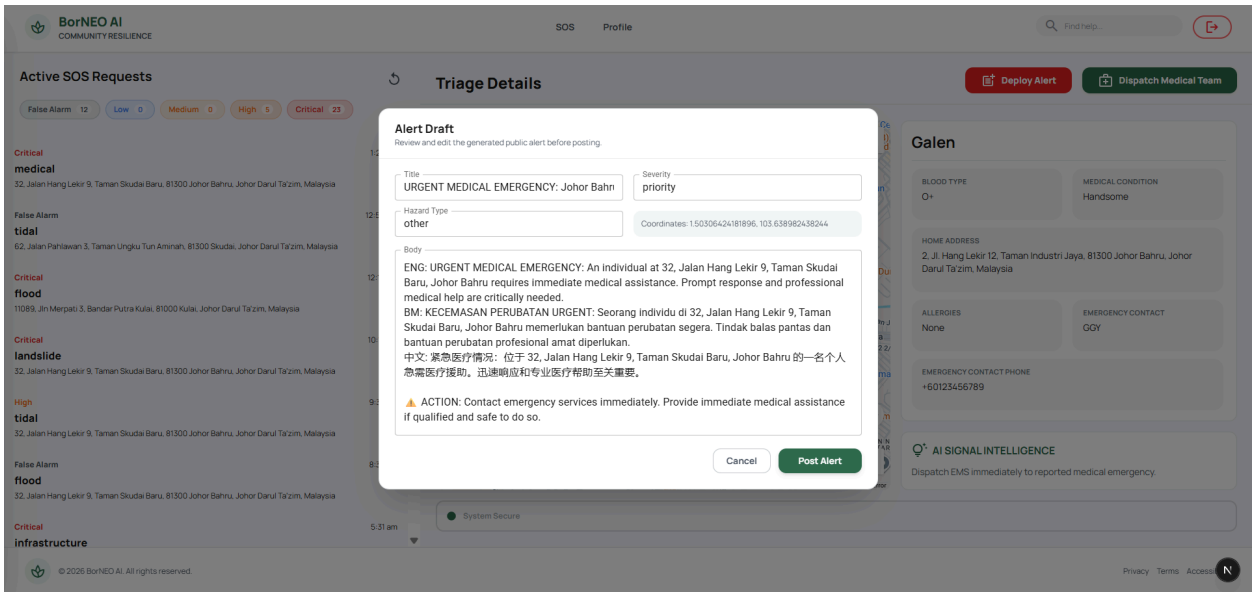




The profile page shows user identity, region, and role. An editable rescue card section captures blood type, allergies, medical conditions, emergency contacts, and home location (with a Google Maps picker). A QR code and shareable URL are generated for field use.

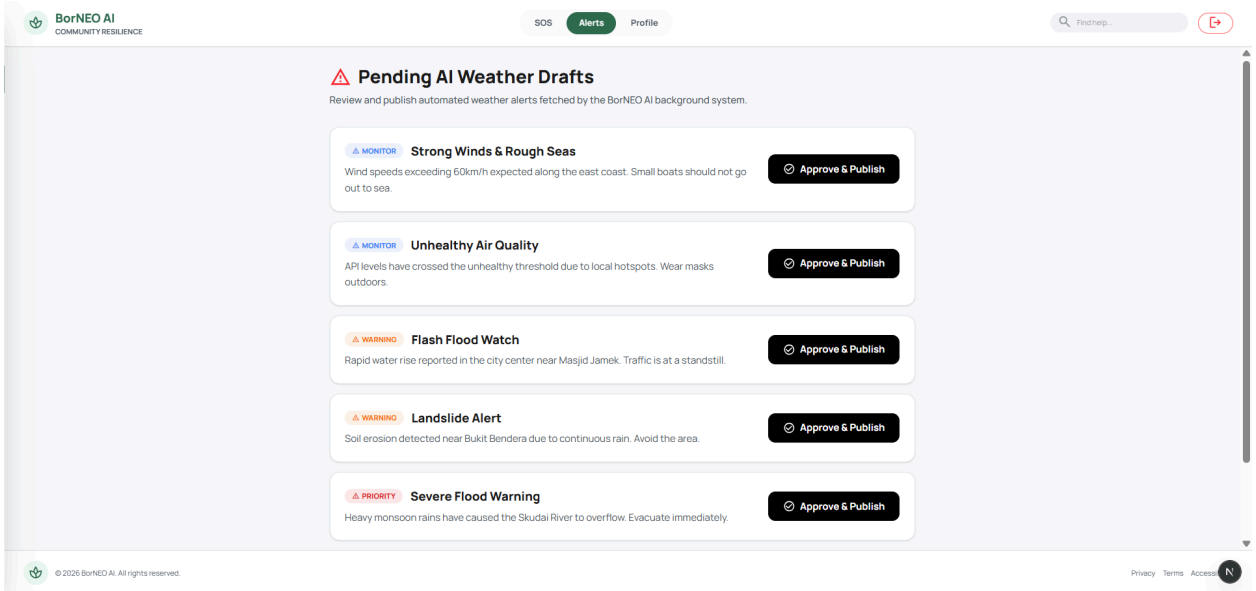
Screenshot 9: Admin SOS Dashboard





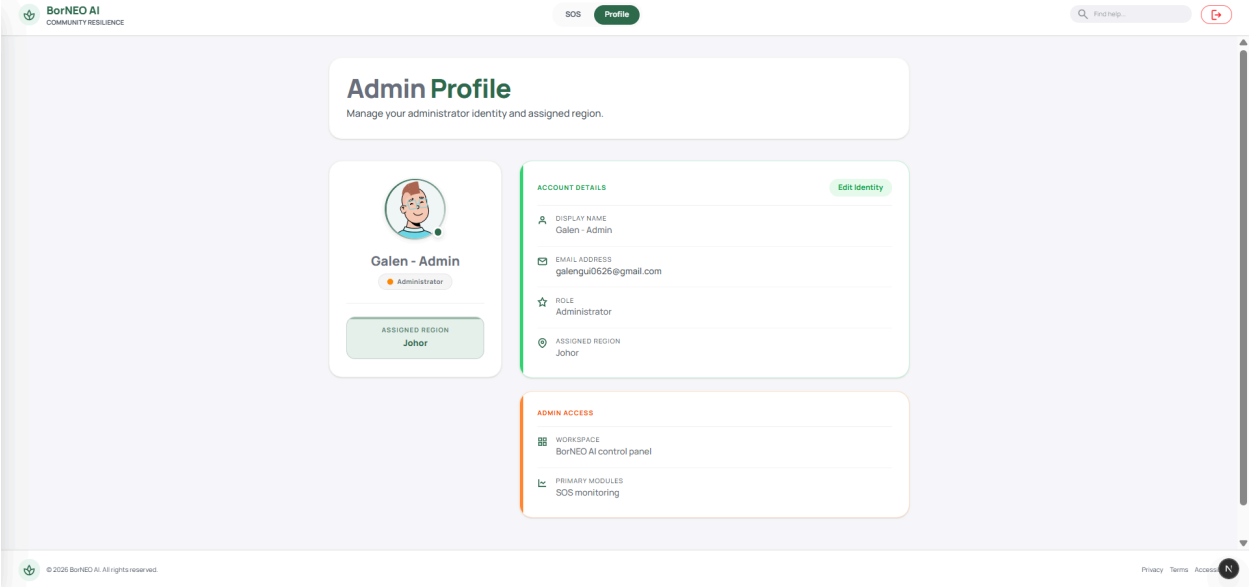
A split-panel layout: the left sidebar lists active SOS requests with severity badges and hazard types; clicking a request shows its detailed view with map, rescue-card data, and AI Signal Intelligence (dispatch recommendation, severity classification, required assets).

Screenshot 10: Admin Alert Broadcasting



The Admin Alert Broadcasting dashboard allows administrators to review AI-generated weather alert drafts before publishing them to the community. Each alert card displays the hazard title, severity level, and a brief description of the situation. Administrators can quickly approve and release the alerts using the **Approve & Publish** button, ensuring timely dissemination of important weather warnings and safety information.

Screenshot 11: Admin Profile



The Admin Profile dashboard allows administrators to view and manage their identity and assigned region. It displays the administrator's avatar, display name, email, role, and region, with an Edit Identity option for profile updates. The Admin Access section also indicates the administrator's workspace and primary module, SOS monitoring, for overseeing emergency alerts.

6. AI Acknowledgement

Disclosure

AI Tool	How It Was Used
GitHub Copilot	Used for code assistance throughout development — implementing features, debugging TypeScript errors, refactoring UI components (e.g., SOS modal hazard options, emergency details modal sizing), and writing API route logic.
Google Gemini 2.5 Flash	Integrated as a production feature within four API endpoints: (1) SOS report triage and severity classification. (2) Alert draft generation with trilingual output. (3) Personalized emergency checklist generation. (4) Survival quiz scenario generation. All Gemini calls use structured JSON output mode (responseMimeType: "application/json") for deterministic parsing.
ChatGPT	Used for brainstorming architecture decisions, refining AI prompt engineering for Gemini integrations, and general debugging support during development.

7. Conclusion

Summary

BorNEO AI demonstrates that a single, integrated AI-driven platform can meaningfully address the fragmented disaster response landscape in Borneo and the broader ASEAN region. By combining AI-assisted SOS triage with precise GPS coordinates, digital medical rescue cards with QR codes, AI-driven e-learning through survival simulations and personalised checklists, and role-based workflows for residents and responders, the platform reduces the critical delay between a distress signal and an informed response action.
The solution directly addresses the ASEAN Capacity Building Roadmap 2025–2030's priorities: strengthening local-level preparedness through AI-powered education, improving the mapping of vulnerable infrastructure and households through coordinate capture and geocoding, and moving toward a proactive culture of safety through hyper-local early warning systems. All three operational pillars — Prepare, Monitor, and Respond — are implemented within a modern, production-ready architecture.

Potential Future Improvements

1. AI-powered landslide detection and flood monitoring - Integrate satellite imagery analysis and IoT sensor data to provide proactive hazard detection before incidents occur, as outlined in the ASEAN Roadmap's vision for AI-driven risk monitoring.
2. Offline-first capability - Implement service workers and local caching so residents can access rescue cards, checklists, and cached alerts without connectivity — critical for rural Borneo during disasters.
3. SMS/USSD fallback - Provide a low-bandwidth SOS submission channel for users without smartphone data access, extending coverage to the most vulnerable populations.
4. School-aged population modules - Develop age-appropriate disaster preparedness simulations and curriculum-aligned content for schools across ASEAN Member States.
5. Multi-agency integration - Connect with official disaster agencies (NADMA, APM, BOMBA, AHA Centre) for direct alert ingestion and dispatch handoff.
6. Real-time SOS tracking - Add WebSocket-based live location streaming so responders can track a resident's movement after an SOS is sent.
7. Analytics dashboard - Build historical incident analytics for administrators to identify high-risk zones, response time trends, and preparedness gaps across regions.

References

1. ASEAN Capacity Building Roadmap 2025–2030 on Disaster Management. AHA Centre. https://ahacentre.org/wp-content/uploads/2024/10/ASEAN-Capacity-Building-Roadmap-2025-2030-on-Disaster-Management-Spread-Version.pdf
2. How AI is being used to respond to natural disasters in cities. Disaster Risk Reduction Knowledge Service. https://ikcest-drr.data.ac.cn/post/1fa34

8. Appendix

1. Website URL: https://borneo-hackathon-web-beige.vercel.app/

2. Github URL: https://github.com/ACZH05/borneo-hackathon

3. Demo Video: https://youtu.be/FFtnrWynUGQ